

PSP - Preprod Monolith Database Migration Tiptoe Performance Analysis

Observations:

- Queries coming from EmailGateway going for full table scan (3cqb8y572aqjv, 5jzvumvvxpv5)
- Noticed query on PSP_FINANCIAL_TRANS_STATE table without company_fk filter (2gnvs7a6qtc dz)
- Noticed query on PSP_FINANCIAL_TRANSACTION table without company_fk filter (50yjqp3zgabva, 27xxc4nytp1a)
- Noticed query on PSP_COMPANY_EVENT_DETAIL table without company_fk filter (fa3nx66gdftdg)

SQL ordered by Elapsed Time

Elapsed Time (s)	Executions	Elapsed Time per Exec (s)	% Total	% CPU	% IO	SQL Id	SQL Module	SQL Text
818.51	1	818.51	17.51	20.76	80.94	c9m6rzckuum9h	JDBC Thin Client	-- assisted_agency_info_rpt_Sl...
702.27	8	87.78	15.03	41.31	58.98	c25bb1q9hyft6	EmailGateway:50	/* load one-to-many com.intuit...
291.38	14	20.81	6.23	4.04	97.53	d7vdj7d9c6adq	EagerLoadTest:497	/* load one-to-many com.intuit...
268.39	30	8.95	5.74	9.96	91.92	2n44a8qv28fbs	TransactionsWS:3915	select financialt0_FINANCIAL_...
261.51	1	261.51	5.60	42.98	57.98	0pjy6qqxz9r3k	JDBC Thin Client	SELECT FED_TAX_ID_ENC FED_TAX_...
213.96	88	2.43	4.58	28.43	74.04	42qtb80mqf0pp	TransactionsWS:3915	select moneymovem0_ MONEY_MOVE...
203.72	2	101.86	4.36	37.48	62.88	3cqb8y572aqjv	EmailGateway:50	/* load one-to-many com.intuit...
180.27	7	25.75	3.86	8.91	94.57	2gnvs7a6qtc dz	TransactionsWS:658	/* FinancialTransaction:getFin...
124.87	24	5.20	2.67	17.34	85.67	50yjqp3zgabva	TransactionsWS:215	/* BaseMoneyMovementTransactio...
115.33	42	2.75	2.47	99.02	0.96	7414941kx90vu	EntitlementActivation:55	/* EntitlementActivation:execu...
99.24	1	99.24	2.12	30.65	69.73	fa3nx66gdftdg	EmailGateway:50	/* load one-to-many com.intuit...
98.67	1	98.67	2.11	31.48	69.00	5jzvumvvxpv5	EmailGateway:50	/* load one-to-many com.intuit...
97.93	1	97.93	2.10	35.77	64.68	52jgky5ka0rss	EmailGateway:50	/* load one-to-many com.intuit...
63.98	2	31.99	1.37	4.30	97.76	27xxc4nytp1a	PSPQueryFinancialTransaction s:98	/* load one-to-many com.intuit...
52.91	24	2.20	1.13	14.59	86.94	2tat9ytdadsa	TransactionsWS:3915	/* load one-to-many com.intuit...

SQL ordered by Reads

Physical Reads	Executions	Reads per Exec	% Total	Elapsed Time (s)	% CPU	% IO	SQL Id	SQL Module	SQL Text
50,536,397	8	6,317,049.63	55.93	702.27	41.31	58.98	c25bb1q9hyft6	EmailGateway:50	/* load one-to-many com.intuit...
14,300,192	2	7,150,096.00	15.83	203.72	37.48	62.88	3cqb8y572aqjv	EmailGateway:50	/* load one-to-many com.intuit...
7,149,968	1	7,149,968.00	7.91	99.24	30.65	69.73	fa3nx66gdftdg	EmailGateway:50	/* load one-to-many com.intuit...
7,149,408	1	7,149,408.00	7.91	97.93	35.77	64.68	52jgky5ka0rss	EmailGateway:50	/* load one-to-many com.intuit...
7,147,328	1	7,147,328.00	7.91	98.67	31.48	69.00	5jzvumvvxpv5	EmailGateway:50	/* load one-to-many com.intuit...
1,112,762	1	1,112,762.00	1.23	818.51	20.76	80.94	c9m6rzckuum9h	JDBC Thin Client	-- assisted_agency_info_rpt_Sl...
641,906	1	641,906.00	0.71	261.51	42.98	57.98	0pjy6qqxz9r3k	JDBC Thin Client	SELECT FED_TAX_ID_ENC FED_TAX_...
395,357	7	56,479.57	0.44	180.27	8.91	94.57	2gnvs7a6qtc dz	TransactionsWS: 658	/* FinancialTransaction:getFin...
294,636	88	3,348.14	0.33	213.96	28.43	74.04	42qtb80mqf0pp	TransactionsWS: 3915	select moneymovem0_ MONEY_MOVE...
292,135	30	9,737.83	0.32	268.39	9.96	91.92	2n44a8qv28fbs	TransactionsWS: 3915	select financialt0_FINANCIAL_...

- sql_id: 5jzvumvvxpv5

```
/* load one-to-many com.intuit.sbd.payroll.psp.domain.CompanyEvent.CompanyEventDetailSet */
select
  companyeve0_.COMPANY_EVENT_FK as company_event_fk11_56_1_,
  companyeve0_.COMPANY_EVENT_DETAIL_SEQ as company_event_deta1_56_1_,
  companyeve0_.COMPANY_EVENT_DETAIL_SEQ as company_event_deta1_56_0_,
  compan yeve0_.VERSION as version2_56_0_,
  companyeve0_.CREATOR_ID as creator_id3_56_0_,
  companyeve0_.CREATED_DATE as created_date4_56_0_,
  companyeve0_.MODIFIER_ID as modifier_id5_56_0_,
  companyeve0_.MODIFIED_DATE as modified_date6_56_0_,
  companyeve0_.REALM_ID as realm_id7_56_0_,
  companyeve0_.EVENT_DETAIL_TY PE_CD as event_detail_type_8_56_0_,
  companyeve0_.VALUE as value9_56_0_,
  companyeve0_.EVENT_DETAIL_SUBTYPE as event_detail_subt10_56_0_,
  companyeve0_.COMPANY_EVENT_FK as company_event_fk11_56_0_,
  companyeve0_.COMPANY_FK as company_fk12_56_0_
from
  PSP_COMPANY_EVENT_DETAIL companyeve0_
where
  companyeve 0_.COMPANY_EVENT_FK in (: 1, : 2, : 3, : 4)
```

Plan hash value: 3622991062

Id	Operation	Name	Rows	Bytes	Cost (%CPU)	Time	Pstart	Pstop
0	SELECT STATEMENT				1936K(100)			
1	PARTITION HASH ALL		11	2882	1936K (1)	00:01:16	1	8
* 2	TABLE ACCESS FULL	PSP_COMPANY_EVENT_DETAIL	11	2882	1936K (1)	00:01:16	1	8

I	SQL_PROFIL	SQL_PLAN_BASELINE	SQL_ID	CHILD_NUMBER	PLAN_HASH_VALUE	EXECUTIONS	EL_PER_EXEC	TO_CHAR
(LAST_ACTIVE_T		LIO	PIO	AVG_CPU_TIME_PER_EXEC	IO_TIME	AVG_ROWS	FIRST_LOAD_TIME	
Y			5jzvumvvxpr5	0	3622991062	1	98.665815	18-AUG 02:
21		7147512	7147328	31.0642	68.0783	12	2023-08-18/02:20:02	

sql_id: c25bb1q9hyft6

```
/* load one-to-many com.intuit.sbd.payroll.psp.domain.CompanyEvent.CompanyEventDetailSet */
select
  companyeve0_.COMPANY_EVENT_FK as company_event_fk11_56_1_,
  companyeve0_.COMPANY_EVENT_DETAIL_SEQ as company_event_deta1_56_1_,
  companyeve0_.COMPANY_EVENT_DETAIL_SEQ as company_event_deta1_56_0_,
  compan yeve0_.VERSION as version2_56_0_,
  companyeve0_.CREATOR_ID as creator_id3_56_0_,
  companyeve0_.CREATED_DATE as created_date4_56_0_,
  companyeve0_.MODIFIER_ID as modifier_id5_56_0_,
  companyeve0_.MODIFIED_DATE as modified_date6_56_0_,
  companyeve0_.REALM_ID as realm_id7_56_0_,
  companyeve0_.EVENT_DETAIL_TY PE_CD as event_detail_type_8_56_0_,
  companyeve0_.VALUE as value9_56_0_,
  companyeve0_.EVENT_DETAIL_SUBTYPE as event_detail_subt10_56_0_,
  companyeve0_.COMPANY_EVENT_FK as company_event_fk11_56_0_,
  companyeve0_.COMPANY_FK as company_fk12_56_0_
from
  PSP_COMPANY_EVENT_DETAIL companyeve0_
where
  companyeve 0_.COMPANY_EVENT_FK in (
    : 1,
    : 2,
    : 3,
```

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)

Plan hash value: 3622991062

Id	Operation	Name	Rows	Bytes	Cost (%CPU)	Time	Pstart	Pstop
0	SELECT STATEMENT				1963K(100)			
1	PARTITION HASH ALL		273	71526	1963K (2)	00:01:17	1	8
* 2	TABLE ACCESS FULL	PSP_COMPANY_EVENT_DETAIL	273	71526	1963K (2)	00:01:17	1	8
I SQL_PROFIL SQL_PLAN_BASELINE SQL_ID CHILD_NUMBER PLAN_HASH_VALUE EXECUTIONS EL_PER_EXEC TO_CHAR								
(LAST_ACTIVE_T LIO PIO AVG_CPU_TIME_PER_EXEC IO_TIME AVG_ROWS FIRST_LOAD_TIME								

Y		c25bb1q9hyft6	0	3622991062	45	127.088659	18-AUG 03:	
03	7144255	7144059	40.413	87.0328	268	2023-08-16/08:25:06		

```
#1
27xxc4nytp1a
/* load one-to-many com.intuit.sbd.payroll.psp.domain.FinancialTransaction.AssociatedTransactionsSet */
select
  associated0_.ORIGINAL_TRANSACTION_FK as original_transact32_125_1_,
  associated0_.FINANCIAL_TRANSACTION_SEQ as financial_transact1_125_1_,
  associated0_.FINANCIAL_TRANSACTION_SEQ as financial_transact1_125_0_,
  associated0_.VERSION as version2_125_0_,
  associated0_.CREATOR_ID as creator_id3_125_0_,
  associated0_.CREATED_DATE as created_date4_125_0_,
  associated0_.MODIFIER_ID as modifier_id5_125_0_,
  associated0_.MODIFIED_DATE as modified_date6_125_0_,
  associated0_.REALM_ID as realm_id7_125_0_,
  associated0_.FINANCIAL_TRANSACTION_AMOUNT as financial_transact8_125_0_,
  associated0_.SETTLEMENT_DATE as settlement_date9_125_0_,
  associated0_.SETTLEMENT_TYPE_CD as settlement_type_c10_125_0_,
  associated0_.CREDIT_BANK_ACCOUNT_TYPE as credit_bank_accou1_1_125_0_,
  associated0_.DEBIT_BANK_ACCOUNT_TYPE as debit_bank_accoun12_125_0_,
  associated0_.ON_HOLD as on_hold13_125_0_,
  associated0_.SKU as sku14_125_0_,
  associated0_.SKU_QUANTITY as sku_quantity15_125_0_,
  associated0_.ORIGINAL_SETTLEMENT_DATE as original_settleme16_125_0_,
```

```

associated0_.REFUND_TYPE as refund_type17_125_0_,
associated0_.BILL_PAYMENT_SPLIT_FK as bill_payment_spli18_125_0_,
associated0_.PAYCHECK_SPLIT_FK as paycheck_split_fk19_125_0_,
associated0_.BILLING_DETAIL_FK as billing_detail_fk20_125_0_,
associated0_.COMP_ADJUST_SUBMISSION_FK as comp_adjust_submi21_125_0_,
associated0_.COMPANY_LAW_FK as company_law_fk22_125_0_,
associated0_.LAW_FK as law_fk23_125_0_,
associated0_.TAX_PENALTY_INTEREST_FK as tax_penalty_inter24_125_0_,
associated0_.CREDIT_BANK_ACCOUNT_FK as credit_bank_accou25_125_0_,
associated0_.DEBIT_BANK_ACCOUNT_FK as debit_bank_accoun26_125_0_,
associated0_.COMPANY_FK as company_fk27_125_0_,
associated0_.RELATABLE_TRANSACTION_FK as relatable_tr ansac28_125_0_,
associated0_.PAYROLL_RUN_FK as payroll_run_fk29_125_0_,
associated0_.TRANSACTION_TYPE_FK as transaction_type_30_125_0_,
associated0_.CURRENT_TRANSACTION_STATE_FK as current_transacti31_125_0_,
associated0_.ORIGINAL_TRANSACTION_FK as original_transact32_125_0_,
associated0_.MONEY_MOVEMENT_TRANSACTION_FK as money_movement_tr33_125_0_
from
PSP_FINANCIAL_TRANSACTION associated0_
where
associated0_.ORIGINAL_TRANSACTION_FK =: 1

```

Plan hash value: 2294458987

```

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| Id | Operation                                | Name                                | Rows | Bytes | Cost (%CPU) |
| Time      | Pstart| Pstop |                                         |                                         |                                         |
-----
|  0 | SELECT STATEMENT                        |                                     |      |      | 283K(100)   |
|  1 | PARTITION HASH ALL                     |                                     |      |      |              |
| 00:00:12 |      1 |    16 |                                     |                                     |              |
|  2 | TABLE ACCESS BY LOCAL INDEX ROWID BATCHED| PSP_FINANCIAL_TRANSACTION          |      |      | 283K (1)    |
| 00:00:12 |      1 |    16 |                                     |                                     |              |
|  3 | INDEX SKIP SCAN                        | PSP_FINANCIALTRANSACTION_FK3      |      |      | 283K (1)    |
| 00:00:12 |      1 |    16 |                                     |                                     |              |
-----

```

```

#2
50yjqp3zgabva
/* BaseMoneyMovementTransaction:getFinancialTransactionCollection:600 */

/* criteria query */
select
this_.FINANCIAL_TRANSACTION_SEQ as financial_transact1_125_0_,
this_.VERSION as version2_125_0_,
this_.CREATOR_ID as creator_id3_125_0_,
this_.CREATED_DATE as created_date4_125_0_,
this_.MODIFIER_ID as modifier_id5_125_0_,
this_.MODIFIED_DATE as modified_date6_125_0_,
this_.REALM_ID as realm_id7_125_0_,
this_.FINANCIAL_TRANSACTION_AMOUNT as financial_transact8_125_0_,
this_.SETTLEMENT_DATE as settlement_date9_125_0_,
this_.SETTLEMENT_TYPE_CD as settlement_type_c10_125_0_,
this_.CREDIT_BANK_ACCOUNT_TYPE as credit_bank_accou11_125_0_,
this_.DEBIT_BANK_ACCOUNT_TYPE as debit_bank_accoun12_125_0_,
this_.ON_HOLD as on_hold13_125_0_,
this_.SKU as sku14_125_0_,
this_.SKU_QUANTITY as sku_quantity15_125_0_,
this_.ORIGINAL_SETTLEMENT_DATE as original_settleme16_125_0_,
this_.REFUND_TYPE as refund_type17_125_0_,
this_.BILL_PAYMENT_SPLIT_FK as bill_payment_spli18_125_0_,
this_.PAYCHECK_SPLIT_FK as paycheck_split_fk19_125_0_,
this_.BILLING_DETAIL_FK as billing_detail_fk20_125_0_,
this_.COMP_ADJUST_SUBMISSION_FK as comp_adjust_submi21_125_0_,
this_.COMPANY_LAW_FK as company_law_fk22_125_0_,
this_.LAW_FK as law_fk23_125_0_,
this_.TAX_PENALTY_INTEREST_FK as tax_penalty_inter24_125_0_,

```

```

this_.CREDIT_BANK_ACCOUNT_FK as credit_bank_accou25_125_0_,
this_.DEBIT_BANK_ACCOUNT_FK as debit_bank_accoun26_125_0_,
this_.COMPANY_FK as company_fk27_125_0_,
this_.RELATABLE_TRANSACTION_FK as relatable_transac28_125_0_,
this_.PAYROLL_RUN_FK as payroll_run_fk29_125_0_,
this_.TRANSACTION_TYPE_FK as transaction_type_30_125_0_,
this_.CURRENT_TRANSACTION_STATE_FK as current_transacti31_125_0_,
this_.ORIGINAL_TRANSACTION_FK as original_transact32_125_0_,
this_.MONEY_MOVEMENT_TRANSACTION_FK as money_movement_tr33_125_0_

```

```

from
  PSP_FINANCIAL_TRANS ACTION this_
where
  (
    this_.MONEY_MOVEMENT_TRANSACTION_FK =: 1
    and this_.SETTLEMENT_DATE >=: 2
  )

```

Plan hash value: 2841201765

Id	Operation	Name	Rows	Bytes	Cost (%)
0	SELECT STATEMENT				326K
1	PARTITION HASH ALL		1	289	326K
2	TABLE ACCESS BY LOCAL INDEX ROWID BATCHED	PSP_FINANCIAL_TRANSACTION	1	289	326K
3	INDEX SKIP SCAN	PSP_FINANCIAL_TRANSACTION_FK10	1		326K

```

#3
2gnvs7a6qtc dz
/* FinancialTransaction:getFinancialTransactionStates:2856 */

```

```

/* criteria query */
select
  this_.FINANCIAL_TRANS_STATE_SEQ as financial_trans_st1_124_0_,
  this_.VERSION as version2_124_0_,
  this_.CREATOR_ID as creator_id3_124_0_,
  this_.CREATED_DATE as created_date4_124_0_,
  this_.MODIFIER_ID as modifier_id5_124_0_,
  this_.MODIFIED_DATE as modified_date6_124_0_,
  this_.REALM_ID as realm_id7_124_0_,
  this_.TRANSACTION_STATE_EFF_DATE as transaction_state_8_124_0_,
  this_.INSERT_USER_ID as insert_user_id9_124_0_,
  this_.GEMS_UPLOAD_BATCH_FK as gems_upload_batch10_124_0_,
  this_.COMPANY_FK as company_fk11_124_0_,
  this_.FINANCIAL_TRANSACTION_FK as financial_transac12_124_0_,
  this_.TRANSACTION_STATE_FK as transaction_state13_124_0_,
  this_.TRANSACTION_RESPONSE_FK as transaction_respo14_124_0_,
  this_.TRANSACTION_TYPE_FK as transaction_type_15_124_0_

```

```

from
  PSP_F INANCIAL_TRANS_STATE this_
where
  (
    this_.FINANCIAL_TRANSACTION_FK =: 1
    and this_.TRANSACTION_STATE_EFF_DATE >: 2
  )

```

Plan hash value: 2721035676

Id	Operation	Name	Rows	Bytes	Cost (%)
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```

-----
| 0 | SELECT STATEMENT | | | | 19141
(100)| | | |
| 1 | PARTITION HASH ALL | | 1 | 226 | 19141
(1)| 00:00:01 | 1 | 16 |
|* 2 | TABLE ACCESS BY LOCAL INDEX ROWID BATCHED| PSP_FINANCIAL_TRANS_STATE | 1 | 226 | 19141
(1)| 00:00:01 | 1 | 16 |
|* 3 | INDEX SKIP SCAN | PSP_FINANCIALTRANSACTIONST_FK2 | 240K| | 2050
(1)| 00:00:01 | 1 | 16 |
-----

```

```

#4
fa3nx66gdftdg
/* load one-to-many com.intuit.sbd.payroll.psp.domain.CompanyEvent.CompanyEventDetailSet */ select companyeve0_.
COMPANY_EVENT_FK as company_event_fk11_56_1_, companyeve0_.COMPANY_EVENT_DETAIL_SEQ as
company_event_deta1_56_1_, companyeve0_.COMPANY_EVENT_DETAIL_SEQ as company_event_deta1_56_0_, compan
yeve0_.VERSION as version2_56_0_, companyeve0_.CREATOR_ID as creator_id3_56_0_, companyeve0_.CREATED_DATE as
created_date4_56_0_, companyeve0_.MODIFIER_ID as modifier_id5_56_0_, companyeve0_.MODIFIED_DATE as
modified_date6_56_0_, companyeve0_.REALM_ID as realm_id7_56_0_, companyeve0_.EVENT_DETAIL_TY
PE_CD as event_detail_type_8_56_0_, companyeve0_.VALUE as value9_56_0_, companyeve0_.EVENT_DETAIL_SUBTYPE as
event_detail_subt10_56_0_, companyeve0_.COMPANY_EVENT_FK as company_event_fk11_56_0_, companyeve0_.COMPANY_FK
as company_fk12_56_0_ from PSP_COMPANY_EVENT_DETAIL companyeve0_ where companyeve
0_.COMPANY_EVENT_FK in (:1 , :2 , :3 )

```

Plan hash value: 3622991062

```

-----
| Id | Operation | Name | Rows | Bytes | Cost (%CPU)| Time | Pstart| Pstop
|-----|-----|-----|-----|-----|-----|-----|-----|
| 0 | SELECT STATEMENT | | | | 1936K(100)| | |
| | | | | | | |
| 1 | PARTITION HASH ALL | | 8 | 2096 | 1936K | (1)| 00:01:16 | 1
| 8 | | | | | | |
|* 2 | TABLE ACCESS FULL| PSP_COMPANY_EVENT_DETAIL | 8 | 2096 | 1936K | (1)| 00:01:16 | 1
| 8 | | | | | | |
-----

```

AWR report for TipToe window



awr_PPSPHP01_1...0200-0300.html